APERIODIC MECHANICAL STIRRING OF A MATERIAL IN A 2D IDEAL FLUID

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Abstract. In 2003, Phillip Boyland's paper, "Dynamics of two-dimensional time-periodic Euler fluid flows", describes how stirring fluids in certain periodic patterns (stirring protocols) will result in the generic vorticity, ω_t, mixing very efficiently (i.e. $\|\nabla\omega_n\|_{C^0} \geq c\lambda^n$, for every $n \in \mathbb{N}$). This paper is concerned with describing a methodology of determining when these results can be replicated in the case that the stirring protocol is not periodic.

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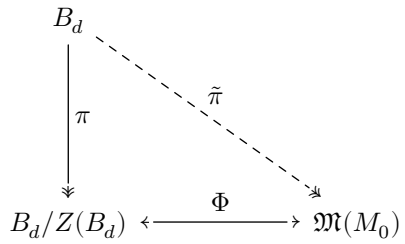
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1. Introduction

The stirring of an ideal fluid by d stirrers in a two dimensional container, M, (meaning an oriented compact surface) can be modeled as a smooth family of diffeomorphisms $\phi_t : M_0 \to M_t$ that depend on $t \geq 0$, where M_t is the container with space occupied by the stirrers carved out of M. Since the stirrers move as a solid (neglecting rotation) through the fluid, they may be traced by looking at a single point's trajectory for each. Denoting these trajectories as $\alpha_1, ..., \alpha_d : [0, \infty) \to M$, it can be further noted that at given time t, $\alpha_i(t) \neq \alpha_j(t)$ if $i \neq j$. So, they may be considered packaged as: $\alpha = (\alpha_1, ..., \alpha_d) : [0, \infty) \to \mathcal{F}_d(M)$, where $\mathcal{F}_d(M) \coloneqq M^d - \{(x_1, ..., x_d) \mid x_i = x_j \text{ for some } i \neq j\}$ is the ordered configuration space of d points on M. There is a quotient map, p, from $\mathcal{F}_d(M)$ to $\mathcal{C}_d(M) \coloneqq \mathcal{F}_d(M)/\mathfrak{S}_d$, where $\mathfrak{S}_d$ is the permutation group of d elements. If there exists a sequence $\{\pi_n\} \subset \mathfrak{S}_d$ such that $\alpha(n) = \pi_n(\alpha(0))$ (where π_n acts on α by permuting its elements), then $\alpha \mid_{[0,n]}$ descends to a loop in $\mathcal{C}_d(M)$. Fixing $p(\alpha(0))$ as the base point, the fundamental group $\pi_1(\mathcal{C}_d(M))$ is called the braid group of d strands on

M. For the specific case of a closed topological disk, it is called Artin's braid group (or, plainly, the braid group) of d strands and is denoted B_d.

If $\alpha(n) = \pi_n(\alpha(0))$, then $M_0 = M_n$, so $\phi_n : M_0 \to M_0$. Generally, call the diffeomorphisms $\varphi : M_0 \to M_0$ and $\psi : M_0 \to M_0$ *isotopic* if there is a smooth family of diffeomorphisms, $H_t : M_0 \to M_0$, such that $H_0 = \varphi$ and $H_1 = \psi$. One might expect a connection between the isotopies of the diffeomorphism and the path homotopies of closed loops on $\mathcal{C}_d(M)$. Indeed, Emil Artin started to develop this idea in [1] and since then it has produced a very fruitful theory between braids and the mapping class group of M_0. The latter, denoted $\mathfrak{M}(M_0)$, is the collection of all isotopy classes of diffeomorphisms on M_0 fixing, setwise, ∂M. You can read more about this theory in [2, 3] and [4]. But importantly, it supports the following diagram:



Where $Z(B_d) \simeq \mathbb{Z}$ is the center of the braid group. By considering isotopy classes of the subgroup of the diffeomorphism group that fix the outer boundary pointwise, we recover an isomorphism between the braid group and the mapping class groups. This can be taken, instead, to be the standard mapping class group of the closed disk with d holes. However, for our purposes, letting points on the outer boundary move freely is more suitable.

By observing diffeomorphisms as they iterate over certain simple arcs (meaning here and through out, a curve that intersects the boundary components) and simple closed curves, we see that sometimes the curves are forced to stretch. These curves will be called *essential* throughout this paper and they are exactly the curves that can't contract to a point nor deform to the boundary (while leaving endpoints, if any, on the boundary). In a sequence of papers between the 1920s and 1940s, Jakob Nielson studied the classification of homeomorphisms on surfaces and their dynamical behavior on curves. These papers went largely unnoticed. In the 1970s, William Thurston independently rediscovered these classifications through his development of measured foliations. The most important classification of homeomorphisms of surfaces is the pseudo-Anosov (pA) classification introduced by Thurston in [5]. They have the property that there are two foliations, transverse to each other, where, upon iteration of the map, lines along one foliation stretch

by a factor, and lines along the other foliation shrink by the inverse of that factor. We may associate each braid a classification from the mapping classes, such as the pA class. This gives various combinatorial tools to study the dynamics of $\mathfrak{M}(M_0)$. For a historical and technical account of these developments, read [3] and [4].

In 2000, Phillip Boyland, Hassan Aref, and Mark Stremler investigated the braiding of trajectories of periodic point vortex systems on the plane, cylinder, and torus [6]. They found that all regimes on the plane do not exhibit pA behavior, but some regimes on the cylinder *can* display pA behavior. This provided motivation to investigate general applications of the theory of pA behavior to incompressible 2d fluids. In 2003, Phillip Boyland demonstrated that under certain conditions, the uniform norm of the gradient of the vorticity of an ideal fluid, stirred by a pA stirring protocol, will grow at least exponentially [7]. This paper centers on the work done in that paper. To see developments since that paper, read [8].

Particularly for this paper, the goal is to write down an extension of the main result in [7], so that it can be utilized in situations when the stirring protocol, $\alpha : [0,\infty) \to \mathcal{F}_d(M)$, fails to satisify $\left[\alpha \mid_{[0,n]}\right] = \beta^n$, where $\beta \in B_d$ is of the pA class. The main strategy is to adjust the stirring and fluid so that we can apply the tools from the theory of mapping class groups. In Section 2, we discuss what to expect from adjusting the stirring protocol and other details relevant to the adjustment of the fluid. In Section 3, the actual adjustment of the fluid is discussed. To complete the argument, in Section 4, we retrace some of Phillip Boyland's steps in [7], with the prior sections in mind.

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2. Adjustments To A Stirring Protocol

Beyond this point, we consider $M = \mathbb{D}$, the closed unit disk, and fix $d \in \mathbb{N}$. Let $\alpha : [0,\infty] \to \mathcal{F}_d(M)$ be an arbitrary stirring protocol. For convenience, we will denote $\varphi(x) = (\varphi(x_1), ..., \varphi(x_d))$ as the action of a diffeomorphism φ on a point $x = (x_1, ..., x_d) \in \mathcal{F}_d(M)$. We will consider the stirrers to be d disks of radius ε centered at each α_i, where, as before, $\alpha = (\alpha_1, ..., \alpha_d)$. This imposes the condition on α that for all time t,

$$|\alpha_i(t) - \alpha_j(t)| > 2\varepsilon \tag{1}$$

for every $i \neq j$. We want to adjust α until we get a new stirring protocol, $\alpha' : [0,\infty) \to \mathcal{F}_d(M)$, that still satisifies eq. 1, but in addition gives for each $n \in \mathbb{N}$,

$$\left[\alpha' \mid_{[0,n]}\right] = \beta^n \tag{2}$$

for some $\beta \in B_d$. Since $B_d = \pi_1(\mathcal{C}_d(M), \alpha(0))$, the requirement that $\alpha' \,|_{[0,n]}$ is a closed loop after descending to $\mathcal{C}_d(M)$ becomes

$$\alpha'(n) = \pi^n(\alpha(0)) \tag{3}$$

for some $\pi \in \mathfrak{S}_d$. More specifically $\pi = \kappa(\beta)$, where $\kappa : B_d \to \mathfrak{S}_d$ is a surjective group homomorphism defined in the following manner: Let $\sigma_1, ..., \sigma_{d-1}$ be the standard generators of B_d, where σ_i can be viewed as the ith strand crossing over the $(i+1)$th strand of the braid. κ sends σ_i to the transposition $(i(i+1)) \in \mathfrak{S}_d$.

To construct α', we first consider a family of homotopies $H^n = (H_1^n, ..., H_d^n) : [0, n] \times [0, 1] \to \mathcal{F}_d(M)$ that satisfy the following:

$$H^n(t, 0) = \alpha(t) \tag{4.1}$$

$$H^n(0, s) = \alpha(0) \tag{4.2}$$

$$[H^n(\cdot, 1)] = \beta^n \tag{4.3}$$

$$|H_i^n(t, s) - H_j^n(t, s)| \geq 2\varepsilon \tag{4.4}$$

for every $s \in [0, 1]$, $t \in [0, n]$, and $i \neq j$. At worst, the homotopies contract each $\alpha|_{[0,n]}$ to the point $\alpha(0)$, then reconstruct a loop that descends to a loop representing β^n. For example: $H_n : [0, n] \times [0, 1] \to \mathcal{F}_d(M_0)$, as:

$$H_n(t, s) := \begin{cases} \alpha((1 - 2s)t) \text{ if } s \leq \frac{1}{2} \\ \nu_n((2s - 1)t) \text{ if } s > \frac{1}{2} \end{cases} \tag{5}$$

for $s \in [0, 1], t \in [0, n]$, $[\nu_n] = \beta^n \in B_d$. This immediately tells us that we can construct a family of homotopies that can adjust the stirring protocol to whatever we want at any checkpoint time $n \in \mathbb{N}$.

We will see that with a family of homotopies like this, we can construct a homotopy, $H : [0, \infty) \times [0, 1]$, such that $H(t, 0) = \alpha(t)$ and $H(t, 1) = \alpha'(t)$. The construction of this will be postponed to the next section, as it comes for free during the adjustment of the fluid flow.

The family of homotopies can get as arbitrarily complex as we want. Indeed, we can imagine a homotopy that produces as many crossings as we want, as long as it immediately destroys the crossing with the inverse of it. Moreover, the family of homotopies isn't quite tangible to work with. It will be useful to reduce the discourse to a family of paths (or even better, a family of homotopy classes of paths). The following lemma establishes this.

Lemma 2.1. *Suppose $\{H_1^n\}$ and $\{H_2^n\}$ are both families of homotopies that satisfy, eq. 4.1 - eq. 4.4, and, in addition, $H_1^n(n, s) = H_2^n(n, s)$ for each $s \in [0, 1]$ and $n \in \mathbb{N}$. Then $[H_1^n(\cdot, 1)] = [H_2^n(\cdot, 1)] = \beta^n \in B_d$*

Proof. The proof proceeds by constructing the homotopies in $\mathcal{F}_d(M)$ and letting it descend to $\mathcal{C}_d(M)$. First, set

$$K(t,s) \coloneqq \begin{cases} H_1^n(2t,s) \text{ if } t \leq \frac{n}{2} \\ H_2^n(2(n-t),s) \text{ if } t > \frac{n}{2} \end{cases} \tag{6}$$

Since $H_1^n(n,s) = H_2^n(n,s)$ for each $s \in [0,1]$, K will be a continuous homotopy, indicating that $[H_1^n(\cdot,1)] \cdot [H_2^n(\cdot,1)]^{-1} = [K(\cdot,1)] \in B_d$, where the equality comes from `eq. 4.1` - `eq. 4.3`. Finally, by defining

$$\tilde{K}(t,s) = \begin{cases} \alpha(2st) \text{ if } s \leq \frac{1}{2}, t \leq \frac{n}{2} \\ \alpha(2s(n-t)) \text{ if } s \leq \frac{1}{2}, t > \frac{n}{2} \\ K(t, 2s-1) \text{ if } s > \frac{1}{2} \end{cases} \tag{7}$$

It becomes clear that $[K(\cdot,1)] = e \in B_d$ and so $[H_1^n(\cdot,1)] = [H_2^n(\cdot,1)]$ □

With the above lemma in mind, it will become relevant to keep track of the curve that intersects the $t = n$ level of each H^n. We may reparameterize H^n so that

$$\| \frac{d}{ds} H_i^n(n,s) \| \leq 1 \tag{8}$$

for each $i = 1, ..., d$. And then let T_n be the upper bound of the interval that $H^n(t,\cdot)$ is defined on after reparameterization. The behavior of T_n will also be important in the upcoming sections. For convenience, denote

$$\gamma^n(s) = H^n(n,s) = (\gamma_1^n(s), ..., \gamma_d^n(s)) \tag{9}$$

for each $n \in \mathbb{N}$ and $s \in [0, T_n]$.

We know that we can construct a family of homotopies deforming $\alpha|_{[0,n]}$ to braid however we want, but this comes at the cost of losing information about the material being stirred. It will become clear below that the growth of T_n directly corresponds to this loss.

3. Correcting Aperiodic Stirring

3.1. **Ideal Fluids In A Container Being Stirred.**

Before we continue, we must discuss what we mean by "ideal" fluid here. In general, a fluid is ideal if its flow, $\phi_t : M \to M$, is generated by a velocity field $v_t(\phi_t(x)) \coloneqq \dot{\phi}_t(x)$ that satisfy the Euler equations:

$$\dot{v_t} + \nabla_{v_t} v_t = -\nabla p_t \tag{10.1}$$

$$\nabla \cdot v_t = 0 \tag{10.2}$$

$$v_t \cdot n = 0 \tag{10.3}$$

where n is a normal vector of ∂M. In the case here, it becomes a bit different because the stirrers act as moving boundaries. So, let $C_0(t), ..., C_d(t)$ be the boundary components of M_t, where M_t was described in Section 1. For simplicity, assume $C_0(t) = \partial M$. The only condition that needs to be changed is the last:

$$v_t \cdot n_i(t) = \dot{\alpha_i}(t) \cdot n_i(t) \tag{11.1}$$

$$v_t \cdot n_0(t) = 0 \tag{11.2}$$

where $n_i(t)$ is a normal vector of $C_i(t)$. Such a velocity field we will call an *Euler solution compatible with the stirring protocol* α *stirring* $C_1, ..., C_d$.

3.2. **Main Lemma.**

With the considerations out of the way, we move onto how to construct the desired α' as well as the corresponding homotopy discussed in the last section. Below is the lemma that lays the ground work for the main result. It states that there is way to adjust the stirring protocol to braid in the desired way, and that it may be extended to adjusting the stirrers and fluid flow. This adjustment, however, is at the sacrifice of the ideal conditions of the fluid. In Section 4, it will turn out that this sacrifice doesn't matter much in getting the result for the original fluid flow.

Lemma 3.2.1. *Let* $\varepsilon > 0$. *Suppose* $\alpha = (\alpha_1, ..., \alpha_d)$ *satisfies eq. 1 and let* $H_n : [0, n] \times [0, T_n] \to \mathcal{F}_d(M)$ *be a family of homotopies that satisfy equations* `eq. 4.1` - `eq. 4.4` *for some* $\beta \in B_d$ *and satisfy eq. 8. If* $0 < \varepsilon_0 < \varepsilon$, *then there exists,* $\{\eta_t : M \to M\}$, *a smooth family of diffeomorphisms such that for every* $n \in \mathbb{N}$ *the following holds:*

- $\eta_n\big(B_{\varepsilon_0}(\alpha_i(n))\big) = B_{\varepsilon_0}\big(\alpha_{\pi^n(i)}(0)\big)$ *for every* $i = 1, ..., d$, $\pi = \kappa(\beta) \in \mathfrak{S}_d$
- $\|d\eta_n\|_{\text{op}} \leq e^{\frac{2T_n}{\varepsilon - \varepsilon_0}}$
- $\big[\alpha' \mid_{[0,n]}\big] = \beta^n$, *where* $\alpha'(t) = \eta_t(\alpha(t))$

Proof. Fix $K \gg 2$. Let $\rho : \mathbb{R} \to [0, 1]$ such that $\text{supp}(\rho) \subset [0, 1]$ and $\rho \equiv 1$ on $\big[\frac{1}{K}, \frac{K-1}{K}\big]$. Then define

$$\chi(r) = \tilde{\chi}\left(1 - \frac{r - \varepsilon_0}{\varepsilon - \varepsilon_0}\right), \text{ where } \tilde{\chi}(s) = \frac{\int_0^s \rho(x)dx}{\int_0^1 \rho(x)dx} \tag{12}$$

χ is identically 1 when $r \leq \varepsilon_0$ and 0 when $r \geq \varepsilon$. In addition,

$$\|\chi'\|_{C^0} = \frac{1}{\varepsilon - \varepsilon_0} \|\tilde{\chi}'\|_{C^0} \leq \frac{1}{\varepsilon - \varepsilon_0} \frac{1}{\int_0^1 \rho(x)dx} \tag{13}$$

Since $\rho \geq \chi_{[\frac{1}{K}, \frac{K-1}{K}]}$,

$$\|\chi'\|_{C^0} \leq \frac{K}{(K-2)(\varepsilon - \varepsilon_0)} \tag{14}$$

Now fix $n \in \mathbb{N}$ and take $\gamma^n = (\gamma_1^n, ..., \gamma_d^n)$ to be the same from eq. 9. Let $v_t^n = \sum_{i=1}^d \chi(\|x - \gamma_i^n(t)\|)\dot{\gamma}_i^n(t)$. The flow of this vector field, η_t^n, satisfies $\eta_t^n(B_{\varepsilon_0}(\alpha_i(n)) = B_{\varepsilon_0}\big(\alpha_{\pi^n(i)}(0)\big)$. Fix $i = 1, ..., d$, if $\varepsilon_0 < \|x - \gamma_i^n(t)\| < \varepsilon$ and $w \in T_xM$ such that $\|w\| = 1$, then

$$\|dv_t^n \cdot w\| = \|\dot{\gamma}_i^n(t)\| \cdot |\chi'(\|x - \gamma_i^n(t)\|)| \cdot |(\nabla \|x - \gamma_i^n(t)\|) \cdot w| \tag{15}$$

Since $K \gg 2$ and by the Cauchy-Schwarz inequality and eq. 14,

$$\|dv_t^n \cdot w\| \leq \frac{K}{(K-2)(\varepsilon-\varepsilon_0)} \leq \frac{2}{\varepsilon-\varepsilon_0} \tag{16}$$

By Grönwall's inequality, since $\|\frac{d}{\mathrm{dt}} d\eta_t^n \cdot w\| \leq \|dv_t^n\|_{\mathrm{op}} \cdot \|d\eta_t^n \cdot w\|$,

$$\|d\eta_{T_n}^n\|_{\mathrm{op}} \leq e^{\int_0^{T_n} \frac{2}{\varepsilon-\varepsilon_0} ds} = e^{\frac{2T_n}{\varepsilon-\varepsilon_0}} \tag{17}$$

Let $\tau : [0,\infty) \to [0,\infty)$ be a smooth function such that $\mathrm{supp}(\tau) \subset \bigcup_{n=1}^{\infty} \left[n-\frac{1}{3}, n+\frac{1}{3}\right]$, $\tau(n) = T_n$, and $\tau\left(\left[n-\frac{1}{2}, n+\frac{1}{2}\right]\right) = [0, T_n]$. Define $\eta_t = \eta_{\tau(t)}^n$ if $t \in \left[n-\frac{1}{2}, n+\frac{1}{2}\right)$ and $\eta_t = \mathrm{Id}_M$ if $t \in \left[0, \frac{1}{2}\right)$.

Set $H : [0,\infty) \times [0,1] \to \mathcal{F}_d(M)$ to be $H(t,s) = \eta_{s\tau(t)}^n(\alpha(t))$ if $t \in \left[n+\frac{1}{2}, n+\frac{1}{2}\right)$ for $n \in \mathbb{N}$ and $H(t,s) = \alpha(t)$ if $t \in \left[0, \frac{1}{2}\right)$. $H(t,1) = \alpha'(t)$, $H(t,0) = \alpha(t)$, and $\gamma^n\left(T_n \cdot \tau\left(\frac{s-1}{2} + n\right)\right) = H(n,s)$, so, by applying Lemma 2.1, $\left[\alpha'|_{[0,n]}\right] = \left[H(\cdot, 1)\,|_{[0,n]}\right] = [H^n(\cdot, T_n)] = \beta^n$. □

In the last leg of the proof of Lemma 3.2.1, the homotopy H was introduced deforming from α to α'. α' from the statement of the lemma and H are the stirring protocol and homotopy alluded to in Section 2. We will see in the next section that the upper bound provided in the statement of Lemma 3.2.1 will play a major role for getting the lower bound of the curve growth of an essential curve. The larger the upper bound is, the more information is lost about the lower bound. For the rest of the paper, we are concerned with the case of when $\{T_n\}$ is a bounded sequence. However, this can be loosened to let T_n grow unbounded as long as it does so slowly.

4. Lower Bounds On The Mixing Of The Material In The Fluid

4.1. **Pseuodo-Anosov Maps.**

The following theorem is a part of a landmark result in the theory of classifying homeomorphisms on surfaces made by William Thurston in [5]. It is written in terms of $\mathcal{L}$, which is defined as

$$\mathcal{L}(\gamma) = \inf\{l(\gamma') \mid \gamma' \text{ is isotopic to } \gamma\} \tag{18}$$

where γ is a simple curve in M_0 and isotopic here means there exists a homotopy, $H : N \times [0,1] \to M_0$, between γ and γ' that has that $H(\cdot, s)$ is an embedding of the one-dimensional manifold N into M_0 with boundary points of N (if any) are sent to boundary points of M_0.

Theorem 4.1.1. *If $\varphi : M_0 \to M_0$ is a diffeomorphism belonging to a pA class, then*

$$\mathcal{L}(\varphi^n \circ \gamma) \geq k\lambda^n \text{ for every } n > 0 \tag{19}$$

where γ is an essential simple curve, $k > 0$ depends only on the isotopy type of γ, and λ is the expansion constant of the pA class that φ belongs to.

Since the proof of this theorem will take us too far off the beaten path, we will not prove it here. However, modern proofs of the theorem can be found in [3] and [4].

At this point, it should be noted that when $d < 3$, the disk with d punctures doesn't admit a mapping class that is pA. So from this point, we assume $d \geq 3$.

4.2. **Simple Morse Functions with Essential Regular Sets.**

Recall, our original goal was to study passively transported scalar material in a fluid. The main result relies on Theorem 4.1.1, and it requires the existence of an essential simple curve to be one of the level sets of the scalar material. However, it may be the case that there is no such curve as a level set. It may even be that the typical case has no such curve. Luckily, we know that their existence is typical.

Definition 4.2.2. *Suppose $f \in C^\infty(M_0))$ is a Morse function. f is said to be* simple *if the following are true:*

- *No critical points lie on the boundary*
- *Level sets contain at most one critical point*
- *$f\,|_{\partial M_0}$ is a Morse function on ∂M_0*

Define G to be the set of simple morse functions with no critical level set tangent to a boundary component. The tangency condition implies any critical level set intersecting the boundary will either have two legs and a loop joined at the critical point or have four legs all joined at the critical point. We will call these type II and III critical sets, respectively. We will call local extrema, type I. The final type, type IV, looks like two loops joined at the critical point. The following is a lemma from [7]. The proof below is a reorganized version of the one from that paper.

Lemma 4.2.3. *Let M_0 be the disk with d holes, where $d \geq 3$. For every $2 \leq k \leq \infty$, $G \subset C^k(M_0, \mathbb{R})$ is a dense, open set where each $\mu \in G$ has an essential regular set.*

Proof. The set of simple Morse functions are open and dense in $C^k(M_0, \mathbb{R})$ (see [9]). By studying the neighborhoods around critical points of simple Morse functions and perturbing them, we find that G is dense. Similarly, by studying neighborhoods around critical points of an element of G, we easily see that small enough perturbations remain in G, giving G as open.

Fix $\mu \in G$. Let $C_0, ..., C_d$ be the boundary components of M_0, with $C_0 = \partial M$. We first assume for each C_i, there is a regular closed curve, B_i, that can deform to that component. We may then construct a connected subset $M' \subset M_0$ whose boundary components are B_i. μ restricted to M' has critical sets that look only like type I and type IV. Let E_i be the union of regular level sets that can deform to B_i. One of the boundary components of E_i must be a loop that joins with another to form a critical set of type IV. The other loop can't deform to C_i or deform to the boundary of a topological disk in the interior of M', since then there would be

a nearby regular closed curve outside of E_i that deforms to C_i. So then the other loop deforms to a different boundary component or another type IV critical set. In the former, there is an essential regular closed curve nearby. In the latter, we may focus on another boundary component and repeat the process. At least one of them will be of the former case. We conclude that if each boundary component has a regular closed curve that can deform to it, then there must be an essential regular curve.

Now consider a $\mu \in G$ with no essential regular curve. Then for critical sets of type II and type III, all legs must lie on the same boundary component, otherwise there would be an essential regular arc nearby. Type III critical sets have that any pair of its legs joined at the critical point must be able to contract to the boundary. Type II critical sets must have the loop either able to deform to the same boundary component intersecting the legs or deform to a point. A regular closed curve must be able to deform to a boundary component or to a point. Further, it may be tangent to at most one boundary component, else there will be an essential regular arc nearby. A regular arc must have its two endpoints on the same boundary component, and must be able to deform to that boundary component. Further, it can't be tangent to any other boundary component.

Let F_i be the union of all level sets that intersect the boundary component C_i. F_i is closed since each level set not intersecting C_i will have neighboring level sets that also don't intersect C_i. In addition, $F_i \cap F_j = \varnothing$ when $i \neq j$. If $x \in \partial F_i - C_i$, then notice that it may not be part of a regular arc or critical set of type III. Similarly, it can't be a part of a critical set of type II, where the loop contracts to a point. So it must be a part of a regular closed curve tangent to C_i or a critical set of type II whose loop can deform to C_i. In either case, there is a regular closed curve nearby that can deform to C_i. Doing this for each boundary component, we arrive at a contradiction since the existence of a regular closed curve for each boundary component that can deform to that component implied the existence of an essential regular curve. So the μ we have considered isn't possible. □

4.3. **Main Result.**

Let $\beta \in B_d$ be of the pA class with expansion constant $\lambda > 1$. We are now ready to proceed to the statement and proof of the main result:

Theorem 4.3.4. *Let $\varepsilon > 0$. Suppose $\alpha = (\alpha_1, ..., \alpha_d)$ satisfies eq. 1 and let $H_n : [0, n] \times [0, T_n] \to \mathcal{F}_d(M)$ be a family of homotopies that satisfy equations* `eq. 4.1` *-* `eq. 4.4` *for some $\beta \in B_d$ and satisfy eq. 8. Additionally, suppose $T_n < L$ for $n \in \mathbb{N}$. Fix $\varepsilon > \varepsilon_0 > 0$, if v_t is a compatible Euler solution with the stirring protocol α stirring ε_0-disks and $\omega_0 \in G$ is its initial vorticity (or some passively transported scalar material), then there exists $c > 0$ s.t.*

$$\|\nabla \omega_n\|_{C^0} \geq c\lambda^n \tag{20}$$

Proof. The proof follows the proof of theorem 7 in [7], editing only when necessary. To begin, by Lemma 3.2.1, there exists $\{\eta_t : M \to M\}$ s.t. $\alpha'(t) = \eta_t(\alpha(t))$ corresponds to $\left[\alpha' \mid_{[0,n]}\right] = \beta^n$ for each $n \in \mathbb{N}$. Let ϕ_t be the Euler flow generated by v_t. By construction of η_t, $\eta_n \circ \phi_n : M_0 \to M_0$ and since β represents a pA class, there exists $\lambda > 1$ s.t. if γ is an essential arc or closed curve, then for some $k > 0$,

$$e^{2\frac{L}{\varepsilon - \varepsilon_0}} l(\phi_n \circ \gamma) \geq e^{\frac{2T_n}{\varepsilon - \varepsilon_0}} l(\phi_n \circ \gamma) \geq l(\eta_n \circ \phi_n \circ \gamma) \geq k\lambda^n \tag{21}$$

so

$$l(\phi_n \circ \gamma) \geq \frac{k}{e^{2\frac{L}{\varepsilon - \varepsilon_0}}} \lambda^n \tag{22}$$

Set $D = \frac{k}{e^{2\frac{L}{\varepsilon - \varepsilon_0}}}$. Since $\omega_0 \in G$, there exists either an essential regular closed curve or an essential regular arc that is the connected component of a regular level set of ω_0. We proceed for the case of the arc since the closed curve is proven in the same fashion. Let S_0 be a collection of essential regular arcs parameterized by some $I = [\Omega_L, \Omega_R]$. For any $r \in I$, $K_{0r} \coloneqq \omega_0^{-1}(r) \cap S_0$ is an essential regular arc. Let $S_n \coloneqq \phi_n(S_0)$, and $K_{nr} \coloneqq \phi_n(K_{0r})$.

Notice, there exists some $s \in I$ such that:

$$\min_{r \in I}(l(K_{nr})) = l(K_{ns}) \geq D\lambda^n \tag{23}$$

We have that S_n is equipped with two orthogonal foliations, the gradient lines of ω_n and K_{nr} themselves. We have a diffeomorphism $F : \{(a, r) \mid r \in I, a_1(r) \leq a \leq a_2(r)\} \to S_n$, where $a_1, a_2 : I \to \mathbb{R}$ are smooth and F restricted to $[a_1(r), a_2(r)] \times r$ parameterizes K_{nr} and $F(a \times I)$ is a trajectory of the gradient flow. Since $\nabla\omega_n$ is collinear with $\frac{\partial F}{\partial r}$ and F^{-1} is ω_n on the fibers $a \times I$, we have that

$$1 = \|\frac{\partial}{\partial r}(F^{-1} \circ F)\| = \|\frac{\partial}{\partial r}(\omega_n \circ F)\| = \|\frac{\partial F}{\partial r}\| \, \|\nabla \omega_n\| \tag{24}$$

Since ϕ_t is incompressible,

$$\text{area}(S_0) = \text{area}(S_n) = \int_{\Omega_L}^{\Omega_R} \int_{a_1(r)}^{a_2(r)} |\text{det}F| \; dadr \tag{25}$$

$$= \int_{\Omega_L}^{\Omega_R} \int_{a_1(r)}^{a_2(r)} \|\frac{\partial F}{\partial a}\| \, \|\frac{\partial F}{\partial r}\| \, dadr = \int_{\Omega_L}^{\Omega_R} \int_{a_1(r)}^{a_2(r)} \frac{\|\frac{\partial F}{\partial a}\|}{\|\nabla \omega_n\|} dadr \tag{26}$$

$$\geq \frac{1}{\|\nabla \omega_n\|_{C^0}} \int_{\Omega_L}^{\Omega_R} \int_{a_1(r)}^{a_2(r)} \|\frac{\partial F}{\partial a}\| \, dadr \geq (\Omega_R - \Omega_L) \cdot \frac{l(K_{ns})}{\|\nabla \omega_n\|_{C^0}} \tag{27}$$

Setting $c = \frac{D(\Omega_R - \Omega_L)}{\text{area}(S_0)}$, we get finally

$$\|\nabla \omega_n\|_{C^0} \geq c\lambda^n \tag{28}$$

□

References

1. Artin, E.: Theory of Braids. Annals of Mathematics. 48, 101–126 (1947)
2. Kassel, C., Turaev, V.: Braid Groups. Springer New York (2008)
3. Fathi, A., Poénaru, V., Laudenbach, F.: Thurston's Work on Surfaces (MN-48). Princeton University Press, United States (2021)
4. Farb, B., Margalit, D.: A Primer on Mapping Class Groups (PMS-49). Princeton University Press (2012)
5. Thurston, W.P.: On the geometry and dynamics of diffeomorphisms of surfaces. Bulletin (New Series) of the American Mathematical Society. 19, 417–431 (1988)
6. Boyland, P., Aref, H., Stremler, M.A.: Topological fluid mechanics of stirring. Journal of Fluid Mechanics. 403, 277–304 (2000)
7. Boyland, P.: Dynamics of two-dimensional time-periodic Euler fluid flows. Topology and its Applications. 152, 87–106 (2003)
8. Thiffeault, J.-L.: Braids and Dynamics. Springer Cham (2022)
9. A. Jankowski, E.R.: Functions with non-degenerate critical points on manifolds with boundary. Commentationes Mathematicae. 16, (1972)

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